

Enhancing Network Brute Force Attack Detection: A Hybrid Sampling and Ensemble Learning Approach

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Abstract

Machine-learning-based Network Intrusion Detection Systems (NIDS) are widely used to detect malicious traffic, but their performance can decline when attack classes are severely underrepresented. Previous studies have attempted to mitigate this problem through feature selection, sampling, and ensemble learning. However, these components often fail to adequately

address minority attacks. In datasets such as CICIDS2017, Brute Force traffic remains difficult to detect because repeated authentication attempts can resemble benign activity. This study evaluates a Brute-Force-centered CICIDS2017 pipeline that integrates train-only feature selection, SMOTE-Tomek hybrid sampling, and ensemble learning. The workflow consolidates CICIDS2017 labels into eight classes, limits the dominant Benign class, removes invalid records, selects numerical flow features using ANOVA *SelectKBest* and correlation pruning, and applies resampling only to the training partition. Validation results showed that SMOTE-Tomek achieved the strongest macro F1-score among the sampling strategies, while the Stacking Ensemble achieved the best macro F1-score among classifier candidates. SMOTE-Tomek increased Brute Force training representation from 5,204 to 76,385 samples and reduced the imbalance against DDoS from about 30.30:1 to 2.06:1. On the unseen test set, the selected configuration achieved 99.93% macro accuracy, 86.10% macro precision, 90.22% macro recall, 86.93% macro F1-score, and 97.98% macro ROC-AUC. For Brute Force traffic, it achieved 98.70% precision, 92.66% recall, 95.58% F1-score, and 0.0253% FPR. The findings indicate that SMOTE-Tomek and heterogeneous stacking can improve Brute Force detection under multiclass imbalance.

Keywords

Network Intrusion Detection System, Brute Force Detection, CICIDS2017, SMOTE-Tomek, Ensemble Learning

INTRODUCTION

Internet-based communication is now central to file sharing, remote access, and daily computing, but the same connectivity also expands the attack surface for private and organizational data. Recent threat reports show that cyberattacks continue to grow in both volume and stealth: malware-free detections increased from 62% in 2021 to 71% in 2022, while 75% of attacks used to gain initial access were malware-free in 2023.¹ This trend shows that network defense requires detection mechanisms that can recognize suspicious behavior beyond known malware signatures.

Intrusion Detection Systems (IDS) are security mechanisms that monitor system or network activity and generate alerts when malicious behavior is detected (Singh et al., 2022). IDS can be categorized into Host-based IDS (HIDS) and Network-based IDS (NIDS). HIDS monitors host-level information such as logs and resource usage, while NIDS observes network traffic flows. In this study, NIDS is selected because network-flow monitoring is suitable for detecting attack behavior across communication traffic and can be managed more centrally than

¹CrowdStrike Global Threat Reports, 2023 and 2024.

host-by-host monitoring (Almuhanna & Dardouri, 2025; Irfa'issurur & Josaphat, 2025). Flow-based detection is also practical for machine learning because a bidirectional communication session can be represented by numerical attributes such as packet length, duration, timing, port, and flag information. These attributes do not describe user intent directly, but they provide measurable traces of behavior that can be learned by classifiers.

As network data continues to grow, traditional IDS approaches face difficulty in recognizing new or changing attack behavior. Machine learning and deep learning have therefore been used to improve NIDS because they can learn patterns from traffic data rather than relying only on fixed signatures (Ahmed et al., 2024; Almuhanna & Dardouri, 2025). However, ML-based NIDS still has important weaknesses, including high false-positive rates, difficulty generalizing to imbalanced data, high computational cost from redundant features, and deployment challenges in real-time environments (Al Essa & Bhaya, 2024; Almuhanna & Dardouri, 2025; Alsaffar et al., 2024; Prasad et al., 2025; Yang et al., 2024). These weaknesses are especially important when the target class is rare. A classifier can appear accurate by learning the dominant normal or high-volume attack classes, while still failing to detect attack types that appear in much smaller quantities.

Previous studies have attempted to reduce these weaknesses through feature selection, sampling, and ensemble learning. Feature selection can remove redundant network-flow attributes, hybrid sampling can reduce class imbalance, and ensemble learning can combine diverse classifiers to improve robustness. However, these components are often evaluated separately or without sufficient focus on minority attacks. CICIDS2017 is a widely used NIDS benchmark that includes Brute Force traffic, but its class distribution is highly imbalanced, making minority-attack detection difficult (Harini et al., 2023; Irfa'issurur & Josaphat, 2025; Khan et al., 2024; Le et al., 2024). This imbalance matters because Brute Force behavior is not necessarily represented by extreme traffic volume alone. Repeated authentication attempts may generate patterns that overlap with legitimate access behavior, so a model must learn subtle distinctions from flow timing, service access, packet behavior, and protocol-state indicators.

Hybrid sampling methods such as SMOTE-Tomek address imbalance by combining minority oversampling with majority or boundary cleaning (Khan et al., 2024; Le et al., 2024; Nassreddine et al., 2025). SMOTE can increase minority-class exposure by generating synthetic samples, while Tomek-link cleaning can remove samples near overlapping class boundaries. Nevertheless, Brute Force traffic remains difficult because repeated authentication attempts may resemble legitimate user behavior, leading to false alarms or missed detections when a single classifier is used (Irfa'issurur & Josaphat, 2025; Rihan et al., 2023). Therefore, sampling alone is not sufficient for a complete NIDS pipeline. The model also needs a compact feature space that reduces redundant attributes and a classifier design that can capture complementary patterns across the multiclass traffic distribution.

This study evaluates whether train-only SMOTE-Tomek, selected flow features, and heterogeneous ensemble learning can improve Brute Force detection in a multiclass NIDS

setting. The term train-only is important because feature selection and resampling can otherwise introduce data leakage if validation or test information influences training. In this work, leakage-sensitive steps are fitted only on the training partition, while validation and test records remain unresampled. This design makes the final evaluation more representative of the model’s behavior on unseen traffic.

The main contributions of this study are: (1) an integrated CICIDS2017 pipeline that combines label consolidation, Benign-class limitation, train-only feature selection, and SMOTE-Tomek resampling; (2) a validation-based comparison of sampling methods and classifier models for Brute Force detection; (3) a final evaluation of the selected Stacking Ensemble with SMOTE-Tomek using macro metrics and Brute-Force-specific metrics; and (4) an interpretation of selected flow features in relation to Brute Force behavior. These contributions are intended to clarify not only whether the proposed model performs well, but also why the selected sampling, feature, and ensemble components are appropriate for the target attack class.

LITERATURE REVIEW

As global connectivity grows, Intrusion Detection Systems (IDS) increasingly rely on machine learning to address high-dimensional traffic data, class imbalance, and changing attack behavior. Recent experimental studies typically focus on three areas: feature selection, hybrid sampling, and ensemble learning. These areas are closely related. Feature selection influences which traffic patterns are available to the classifier, sampling determines how strongly minority classes appear during training, and ensemble learning affects how prediction errors are reduced across different decision models. Examining these studies identifies the gap that motivates a hybrid approach for improving the detection of minority attacks such as Brute Force.

Feature Selection and Dimensionality Reduction in NIDS

Feature selection is important in NIDS because redundant flow attributes can increase training cost and weaken generalization (Alsaffar et al., 2024; Khan et al., 2024; Nassreddine et al., 2025; Selvakumar et al., 2025; Yu et al., 2025). Network-flow datasets usually contain multiple measurements that are statistically related, such as duration, packet counts, byte counts, timing values, and flag indicators. If all attributes are retained without screening, classifiers may learn redundant patterns, require more computation, or become more sensitive to noise. Feature selection therefore functions as both a computational and interpretive step: it reduces the input space and helps identify which flow behaviors are most relevant to detection.

Nassreddine et al. (2025) used correlation-based and embedded XGBoost feature selection, retained 10 features, and applied KNN-DT-RF-XGBoost stacking, achieving 99.97% accuracy and 99.93% F1-score on CIC-IDS2017. Alsaffar et al. (2024) proposed MI-Boruta with RF-XGBoost-CatBoost-MLP stacking, reaching 99.92% multiclass accuracy on CICIDS2017.

Khan et al. (2024) used SelectKBest, correlation reduction, SMOTE-Tomek, and conventional classifiers, where Decision Tree and AdaBoost reached 96.37% accuracy and 96.33% F1-score for multiclass classification. These studies show that feature selection is a necessary first stage before modeling high-dimensional traffic data. However, they also indicate that feature selection should not be treated only as a preprocessing formality. In the context of Brute Force detection, selected features need to be meaningful for repeated service access, timing, packet behavior, and TCP-state differences, because the target behavior may overlap with benign login or access flows.

This study follows that direction by combining ANOVA-based univariate selection with correlation pruning. The ANOVA stage identifies features that separate labeled classes, while correlation pruning removes highly redundant attributes after ranking. This design is simpler than wrapper-heavy selection methods and is suitable for a controlled experimental pipeline because it preserves interpretability and reduces the number of features before resampling and ensemble training.

Handling Imbalanced Datasets Using Hybrid Sampling

Hybrid sampling is widely used to increase minority-class exposure while reducing ambiguous or noisy samples (Harini et al., 2023; Prasad et al., 2025; Yang et al., 2024). Imbalance is a persistent difficulty in intrusion detection because benign traffic and high-volume attack classes can dominate the training distribution. When this happens, a classifier can minimize overall error while still under-detecting rare but important attacks. For minority attacks, the main problem is not only the number of samples, but also the limited diversity of observed patterns during training. Oversampling can make the minority class more visible, while cleaning methods can reduce overlap between synthetic or original samples and nearby majority samples.

Hooshmand et al. (2024) proposed SKM-XGB, which combines SMOTE with K-means undersampling before XGBoost classification, achieving 97.49% multiclass detection on UNSW-NB15 and 99.22% on NSL-KDD. Khan et al. (2024) balanced CICIDS2017 using SMOTE-Tomek and reported 96.37% multiclass accuracy with Decision Tree and AdaBoost. Nassreddine et al. (2025) also used SMOTE-Tomek before stacking and obtained 99.97% accuracy and 99.93% F1-score on CIC-IDS2017. These findings motivate the use of SMOTE-Tomek for Brute Force traffic, because the method combines synthetic minority generation with boundary cleaning.

At the same time, sampling can introduce evaluation risk if it is applied before data splitting or if synthetic samples indirectly influence validation and test partitions. For this reason, sampling must be applied only to the training data. This study adopts that principle by keeping validation and test data unresampled. As a result, the evaluation reflects performance on the original class distribution rather than on an artificially balanced validation or test set.

This is particularly important when the target is Brute Force detection, because the operational question is not only whether a model learns from a balanced training set, but whether it can recognize Brute Force samples when they appear in an imbalanced traffic environment.

Ensemble Learning Approaches for Intrusion Detection

Ensemble learning is effective for IDS because diverse classifiers can reduce the weakness of any single learner (Alsubaei, 2025; Emanet et al., 2023; Gu et al., 2025; Pramanick et al., 2025; Wardana et al., 2024). Different models can capture different relationships in network-flow data. Tree-based models are useful for nonlinear feature interactions, boosting models can refine difficult decision boundaries, distance-based methods can respond to local sample structure, and linear meta-learners can combine base predictions in a controlled way. In imbalanced NIDS, this diversity is valuable because minority attacks may not be represented well by a single decision rule.

Farooqi et al. (2024) proposed a voting ensemble of Decision Tree, Random Forest, and XGBoost, achieving 99.98% accuracy on CIC-IDS2017 with a very low false positive rate. Urmi et al. (2024) used RF, XGBoost, and Extra Trees with Logistic Regression stacking, reporting 99.90% accuracy and 99.80% F1-score on CICIDS2017. Al Essa and Bhaya (2024) combined RF, XGBoost, and MLP using voting and stacking, where soft voting reached 99.9% accuracy and 99.8% F1-score on InSDN. These studies justify the selection of Random Forest, Decision Tree, XGBoost, KNN, and Logistic Regression as baseline or ensemble components in this study.

Voting and stacking represent two different ensemble strategies. Soft voting averages class probabilities and can stabilize prediction when the component models are individually reliable. Stacking trains a meta-learner to combine base-model outputs and can therefore learn which base learners are more useful for different decision regions. This study evaluates both strategies because Brute Force detection requires not only high overall correctness, but also balanced minority-class behavior. The comparison helps determine whether probability averaging or meta-learning is more suitable after SMOTE-Tomek resampling.

Research Gap

Prior studies often emphasize either feature selection, sampling, or ensemble learning, while fewer studies report how these components interact for Brute Force detection under a multiclass setting. Rare-class and cross-domain experiments also show that limited attack support and distribution shift remain active NIDS challenges (Wardana et al., 2024; Xu et al., 2025). This gap is important because a model that performs well on aggregate multiclass metrics may still provide limited evidence about a specific minority attack. Brute Force detection requires class-specific interpretation because the operational cost of missed attacks and false

alarms is different from the cost of generic multiclass error.

This study addresses that gap by combining train-only ANOVA feature selection, correlation pruning, SMOTE-Tomek, and two ensemble strategies, with the interpretation centered on Brute Force precision, recall, F1-score, and false positive behavior. The study also reports validation diagnostics for sampling and classifier selection, so the final configuration is not chosen only from test-set performance. In this way, the proposed pipeline links methodological choices to both macro-level multiclass behavior and Brute-Force-specific detection behavior.

Another important gap is the way performance is interpreted. Many IDS studies report very high accuracy, but high accuracy can be difficult to interpret when one or two classes dominate the dataset. For Brute Force detection, the more useful evidence is whether the model can maintain high precision, recall, and F1-score for the target class while preserving acceptable macro-level behavior. This study therefore treats accuracy as only one part of the evaluation and gives stronger attention to macro F1-score, macro FPR, and Brute-Force-specific metrics. This evaluation focus is necessary because the proposed method is intended to improve detection of a specific minority attack rather than only maximize overall multiclass correctness.

METHODOLOGY/DATA AND METHODS

This section describes the workflow used to evaluate Brute Force detection in a multiclass NIDS. As shown in Figure 1, the experiment begins with CICIDS2017 label mapping and Benign-class limitation, followed by data cleaning, numerical feature retention, stratified splitting, train-only feature selection, train-only SMOTE-Tomek resampling, ensemble model development, validation diagnostics, and final test evaluation. The workflow was designed to answer a practical modeling problem: whether Brute Force detection can be improved without allowing validation or test data to influence feature selection or resampling.

All leakage-sensitive steps were fitted only on the training partition, while validation and test data remained unresampled. This design is important because oversampling before splitting can create synthetic samples that are too closely related to validation or test records. Similarly, feature selection fitted on the full dataset can transfer information from the evaluation data into the training process. By fitting feature selection and sampling only within the training workflow, the experiment preserves a clearer separation between model development and final evaluation.

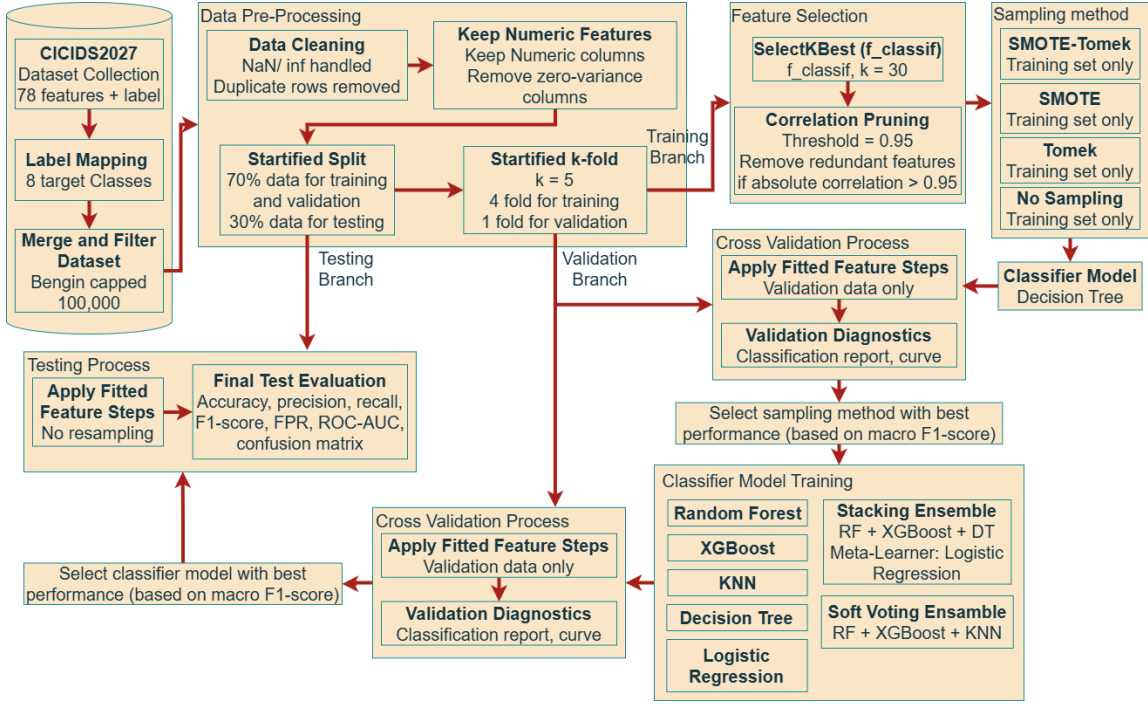


Figure 1. Overall research workflow from dataset preparation to final evaluation.

Source: Authors' processing, 2026.

Dataset

CICIDS2017 is a public benchmark from the Canadian Institute for Cybersecurity, University of New Brunswick, containing 2,830,743 labeled bidirectional flow records with 78 features and one label (Khan et al., 2024; Nassreddine et al., 2025). The raw labels were consolidated into eight categories: BENIGN to Benign; Bot to Bot; DoS labels, DDoS, and Heartbleed to DDoS; PortScan to Port Scan; Infiltration to Infiltrations; Patator and Web Attack-Brute Force labels to Brute Force; XSS labels to XSS; and Sql Injection to SQL-Injection. No labels were dropped after mapping. To reduce normal-traffic dominance while retaining attack samples, Benign was randomly limited to 100,000 records using random seed 42. The resulting class counts are shown in Table 1.

The label consolidation step was used to make the classification task more suitable for Brute-Force-centered multiclass evaluation. Several original CICIDS2017 labels represent related attack behavior, and grouping them into broader categories reduces unnecessary fragmentation while preserving the main attack families. Brute Force was formed from Patator and Web Attack-Brute Force labels because both represent repeated authentication or access attempts. The Benign limitation was applied to reduce the dominance of normal traffic, but attack classes were retained after mapping so that the minority-class structure of the dataset remained visible.

Table 1. Class Distribution Across Main Processing Stages

Label	Cleaned	Train Before	Train After
Benign	97,947	47,994	76,724
Bot	1,953	957	76,996
Brute Force	10,622	5,204	76,385
DDoS	321,775	157,669	157,551
Infiltrations	36	17	77,000
Port Scan	90,819	44,501	76,842
SQL-Injection	21	11	76,999
XSS	652	320	76,400

Note. Train After refers to the training partition after SMOTE-Tomek resampling.

Table 1 shows that the dataset remained strongly imbalanced even after Benign limitation. DDoS remained the largest attack category with 321,775 cleaned records, while Brute Force contained 10,622 cleaned records. Other attack classes were much smaller, including 652 XSS records, 36 Infiltrations records, and 21 SQL-Injection records. This distribution confirms that the experiment was not only a general multiclass classification task, but also a minority-attack detection task. It also justifies the use of macro metrics, because overall accuracy would be dominated by larger classes if each class were not considered separately.

Data Preprocessing

Data preprocessing was performed to improve data quality and model reliability on CICIDS2017. The cleaning stage removed invalid records, redundant entries, and missing values so that the classifiers were not trained on incomplete or duplicated observations. The feature set was restricted to numerical attributes because the selected models require numerical input and because CICIDS2017 flow features are primarily represented as quantitative network measurements. This step also kept the modeling pipeline focused on flow behavior rather than on non-numerical metadata.

To preserve the original class proportions during evaluation, stratified splitting was used

for the train-validation-test partitions. The initial split allocated 49% of the data to training, 30% to testing, and 21% to validation. Stratification was also maintained during five-fold validation so that each fold contained a proportional representation of the available classes. This was necessary because some attack categories, especially SQL-Injection and Infiltrations, contained very few samples after preprocessing. Without stratification, a fold could contain too few minority samples to provide meaningful validation feedback.

The test set was kept separate from the model-selection process. Sampling-method selection, classifier selection, and feature selection decisions were made using training and validation data only. The final selected configuration was then evaluated on the unseen test partition. This separation supports a more reliable interpretation of the results because the final test metrics represent performance after the modeling decisions had already been made.

Feature Selection

ANOVA-based univariate selection (*SelectKBest* with `f_classif`) was implemented to improve interpretability and reduce dimensionality. The selector was fitted only on the training partition to prevent data leakage, ensuring that validation and test data remained unseen during feature ranking. *SelectKBest* with `f_classif` was used to identify the top 30 features because it ranks numerical attributes according to their ability to separate discrete class labels. This is suitable for multiclass intrusion detection because the labels represent distinct traffic categories.

After univariate ranking, Pearson correlation pruning was applied to the selected features. Features were removed when their absolute correlation coefficient exceeded $|r| > 0.95$, because highly correlated attributes can carry similar information and may increase model complexity without adding meaningful discrimination. The process reduced the input space from 70 numerical features to 16 final features. This reduction allowed the ensemble models to focus on a smaller group of discriminative and less redundant variables.

Feature selection was performed before resampling so that synthetic samples were generated in the reduced feature space. This sequence prevents synthetic data from influencing which features are selected. It also reduces the computational burden of SMOTE-Tomek because the sampler operates on fewer dimensions. As a result, the selected features served both methodological and practical purposes: they reduced redundancy and helped make the final model easier to interpret.

Imbalanced Data Handling

To determine the sampling method, this research evaluated four strategies: no sampling, SMOTE, Tomek Links, and SMOTE-Tomek. A Decision Tree classifier was used during this sampling-selection stage so that the sampling methods could be compared under the same classifier setting. No sampling served as the baseline, SMOTE represented oversampling, Tomek

Links represented boundary cleaning, and SMOTE-Tomek represented a hybrid approach.

The selection was based on macro validation metrics because the target problem is imbalanced. Macro metrics give each class equal weight, so they are more informative than accuracy when minority attacks are important. SMOTE increases the number of minority samples by creating synthetic examples, but synthetic samples may appear near regions where minority and majority classes overlap. Tomek Links can remove ambiguous boundary samples, but cleaning alone does not increase minority-class exposure. SMOTE-Tomek was therefore selected because it combines the strengths of both approaches: it increases minority representation and removes overlapping or ambiguous samples after oversampling.

In this experiment, SMOTE-Tomek was applied only to the training data. The validation and test partitions were not resampled, because the goal was to evaluate the model on data distributions that were not artificially balanced. This design is especially relevant for Brute Force traffic, because operational NIDS data are expected to remain imbalanced even if the training procedure uses resampling.

Proposed Ensemble Models

Two ensemble models were trained after sampling-method selection. The soft voting ensemble used Random Forest, XGBoost, and KNN, combining predicted probabilities by averaging with $n_jobs=1$. This approach assumes that the probability estimates from different base learners can complement each other and produce a more stable final prediction. The stacking ensemble used Random Forest, XGBoost, and Decision Tree as base learners, with Logistic Regression as the meta-learner, five-fold internal cross-validation, and $n_jobs=1$. Stacking differs from voting because the meta-learner is trained to combine the outputs of the base learners rather than simply averaging them.

Random Forest used 300 trees, maximum depth 20, balanced class weights, random seed 42, and $n_jobs=-1$. XGBoost used 300 estimators, maximum depth 8, learning rate 0.05, subsample 0.8, column subsample 0.8, histogram tree method, `device=cuda`, `mlogloss`, and random seed 42. Decision Tree used maximum depth 20, balanced class weights, and random seed 42. Logistic Regression used $C = 0.5$, maximum iteration 2000, balanced class weights, random seed 42, and $n_jobs=-1$. KNN used $k = 3$, Euclidean distance, and $n_jobs=-1$. These components were selected because prior NIDS studies reported strong results using tree-based learners, boosting, KNN, voting, and stacking for imbalanced intrusion detection (Farooqi et al., 2024; Nassreddine et al., 2025; Urmi et al., 2024).

The model set was intentionally heterogeneous. Random Forest and Decision Tree provide tree-based decision structures, XGBoost provides boosted tree learning, KNN adds a local distance-based perspective, and Logistic Regression provides a simple meta-learning layer for stacking. This diversity is useful because Brute Force samples may be separated from benign traffic through different types of evidence, including service ports, timing behavior, and

packet-level characteristics.

Experimental Setup

The experiment was run in Google Colab using Python 3 and an NVIDIA L4 GPU. The notebook installed `imbalanced-learn`, `xgboost` $\geq 2.0.0$, `seaborn`, and `joblib` when needed, but exact package versions were not recorded. All stochastic stages used random seed 42 where supported. The use of a fixed random seed improved repeatability for splitting, model initialization, and sampling behavior, although exact reproducibility may still depend on package versions and runtime configuration.

The experimental workflow consisted of two selection stages and one final evaluation stage. First, the sampling methods were compared using validation metrics. Second, classifiers were compared after the selected sampling method was fixed. Finally, the selected Stacking Ensemble with SMOTE-Tomek was trained on the combined train-validation partition and evaluated on the held-out test set. This staged design avoids selecting the final configuration directly from the test set.

Evaluation Metrics

The evaluation used accuracy, precision, recall, per-class F1-score, false positive rate (FPR), macro-averaged F1-score, and macro one-vs-rest ROC-AUC. Accuracy was retained because it is a common classification metric, but it was not treated as the main indicator of minority-class performance. Precision, recall, F1-score, and FPR were emphasized because they describe different operational concerns. Precision indicates how reliable a Brute Force alert is, recall indicates how many actual Brute Force samples are detected, F1-score balances precision and recall, and FPR indicates the rate of false alarms.

Macro-averaged metrics were used to assess class-balanced performance across all classes. This is important in CICIDS2017 because the class distribution remains uneven even after Benign limitation. In the one-vs-rest setting, each class was treated as the positive class in turn, with TP, TN, FP, and FN denoting true positives, true negatives, false positives, and false negatives. The metrics were computed as follows:

1. Accuracy: Accuracy measures the proportion of correct predictions, including true positives and true negatives, to the total number of instances.

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} \quad (1)$$

2. Precision: Precision measures the proportion of correctly predicted attacks to all samples predicted as attacks.

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \quad (2)$$

3. Recall: Recall measures the proportion of correctly predicted positive instances out of all actual positive instances.

$$\text{Recall} = \frac{TP}{TP + FN} \quad (3)$$

4. F1-score: F1-score is the harmonic mean of precision and recall.

$$F1 = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (4)$$

5. False Positive Rate: FPR measures the proportion of negative instances that are incorrectly predicted as positive.

$$\text{FPR} = \frac{FP}{FP + TN} \quad (5)$$

Because accuracy can hide weak minority-class detection, the discussion emphasizes Brute Force precision, recall, F1-score, and FPR.

The metric design also supports fairer interpretation of the sampling and classifier selection stages. During sampling selection, macro metrics were used to compare whether each strategy improved the balance of performance across classes, not merely whether it increased the number of correct predictions. During classifier selection, macro F1-score was used as the main selection indicator because it combines precision and recall under equal class weighting. The final test analysis then reports both macro metrics and Brute-Force-specific metrics so that the result can be interpreted at two levels: overall multiclass robustness and target-attack detection quality.

RESULTS AND DISCUSSION

This section presents the experimental findings as a connected interpretation of class balancing, feature contribution, and classifier selection. Each discussion begins with the observed result, explains its meaning, connects it with previous studies, states its implication, and then clarifies the consequence for Brute Force detection. Macro-averaged metrics are emphasized because accuracy can remain high in imbalanced NIDS data even when minority attack behavior is less reliably detected. For this reason, the interpretation gives more attention to macro F1-score, macro recall, macro FPR, and Brute-Force-specific results than to accuracy alone.

Effect of Hybrid Sampling on Class Representation

Table 2 shows the five-fold validation results used to select the sampling strategy. SMOTE-Tomek achieved the strongest macro F1-score at 83.83%, the highest macro precision at 83.85%, the highest macro recall at 89.10%, and the highest macro ROC-AUC at 94.99%. Tomek Links produced the highest macro accuracy at 99.93% and the lowest macro FPR at

0.0450%, but its macro F1-score was lower than the no-sampling baseline.

Table 2. Sampling Method Selection Based on Validation Macro Performance

Sampling Method	Macro Acc.	Macro Prec.	Macro Rec.	Macro F1	Macro FPR	Macro ROC-AUC
No Sampling	99.92%	80.05%	88.09%	82.02%	0.0484%	94.35%
SMOTE	99.91%	82.86%	88.41%	82.99%	0.0519%	94.60%
SMOTE-Tomek	99.91%	83.85%	89.10%	83.83%	0.0508%	94.99%
Tomek	99.93%	79.65%	86.43%	81.20%	0.0450%	93.42%

Note. Sampling strategies were evaluated with a Decision Tree classifier across five validation folds.

The meaning of this result is that cleaning majority-boundary samples alone can reduce false positives, but it does not necessarily improve class-balanced detection. SMOTE-Tomek was stronger because it combined synthetic minority oversampling with Tomek-link cleaning. In the final training partition, Brute Force increased from 5,204 to 76,385 samples after SMOTE-Tomek. Relative to DDoS, the largest attack class, the Brute Force imbalance changed from approximately 30.30:1 before resampling to 2.06:1 after resampling. This does not make the dataset perfectly uniform, because DDoS remained larger than the other classes, but it substantially reduced the pressure that previously made Brute Force easier to overlook during model fitting.

This finding is consistent with studies showing that hybrid sampling can improve minority-attack representation in imbalanced intrusion datasets (Harini et al., 2023; Khan et al., 2024; Le et al., 2024; Nassreddine et al., 2025). The selection of SMOTE-Tomek is therefore justified because it improved the most important macro-balanced indicators while also increasing the Brute Force training representation substantially. The implication is that resampling should not be judged only by overall accuracy, because the accuracy differences among sampling methods were very small. Consequently, SMOTE-Tomek became the most appropriate sampling method for this pipeline because it gave Brute Force traffic a more meaningful role in model learning without discarding the multiclass structure of the dataset.

The comparison also shows why a hybrid method is preferable to using only one balancing mechanism. SMOTE improved macro F1-score compared with no sampling, but SMOTE-Tomek produced a further improvement while also achieving the strongest macro ROC-AUC among the sampling candidates. Tomek Links alone produced the lowest macro FPR, but its lower macro recall and macro F1-score indicate that cleaning without increasing minority exposure was not sufficient for the target problem. The consequence is that the sampling choice should be interpreted as a balance between representation and boundary control. For Brute Force detection, increasing representation is essential because the original training data contain far fewer Brute Force samples than DDoS samples.

The result also explains why the final discussion does not treat every minority class

as equally improved. SMOTE-Tomek created a much larger training representation for rare classes, but the quality of synthetic expansion still depends on the information available in the original samples. Brute Force had enough original support to benefit from resampling more reliably than SQL-Injection or Infiltrations. Therefore, the sampling result supports the main objective of improving Brute Force detection, while also showing that extremely small classes require more cautious interpretation.

Selected Flow Features and Detection Meaning

The feature-selection process reduced the original numerical feature space to 16 final features through ANOVA-based *SelectKBest* and correlation pruning. Permutation-importance diagnostics showed that Destination Port was the most influential feature. The remaining high-ranked features were grouped into four flow-behavior categories: TCP window behavior, timing behavior, packet behavior, and TCP flag behavior.

The meaning of this feature pattern is behaviorally plausible for Brute Force detection. These features describe service targeting, TCP initial window behavior, inter-arrival timing, connection duration, packet-size variation, traffic rate, and TCP flag behavior. Brute Force activity often concentrates on specific services and produces repeated connection attempts, so destination service information, timing regularity, packet-length behavior, and TCP-state characteristics can become useful indicators.

The result supports previous NIDS studies that use feature selection to reduce redundant flow attributes before classification (Alsaffar et al., 2024; Nassreddine et al., 2025). It also agrees with studies that treat Brute Force and related web attacks as behavior-driven detection problems rather than purely volume-driven anomalies (Irfa'issurur & Josaphat, 2025; Rihan et al., 2023). The use of these selected features is justified because they are not only statistically important, but also meaningful in terms of how Brute Force traffic appears in network flows. The implication is that the selected model did not depend only on opaque statistical separation; it relied on flow features that can be interpreted in terms of network behavior. Consequently, the result strengthens the practical credibility of the pipeline for CICIDS2017 Brute Force detection, although the same feature ranking should be validated on newer or different datasets before being generalized.

This interpretation is important because feature importance alone can be misleading if the selected variables have no clear behavioral connection to the target attack. In this study, the most influential feature, Destination Port, is consistent with the idea that Brute Force activity often targets particular services. Timing and duration features are also meaningful because repeated access attempts can alter the rhythm of a flow. Packet-length and packet-rate features provide additional evidence about how the communication behaves after the connection is initiated. The consequence is that the selected feature set supports both performance and interpretability: it reduces the input space while still retaining variables that can be explained

in terms of network behavior.

Classifier Selection and Minority-Class Robustness

Table 3 reports the classifier-selection results after SMOTE-Tomek was chosen as the sampling method. The Stacking Ensemble achieved the highest validation macro F1-score at 84.78%, the highest macro precision at 86.35%, the highest macro accuracy at 99.93%, and the lowest macro FPR at 0.0406%. XGBoost achieved slightly higher macro recall at 89.27% and the highest macro ROC-AUC at 99.98%, while Soft Voting also produced a very high macro ROC-AUC at 99.98%.

Table 3. Classifier Model Selection Based on Validation Macro Performance

Classifier Model	Macro Acc.	Macro Prec.	Macro Rec.	Macro F1	Macro FPR	Macro ROC-AUC
Decision Tree	99.91%	83.85%	89.10%	83.83%	0.0508%	94.99%
KNN	99.61%	63.54%	80.68%	67.07%	0.2719%	90.83%
Logistic Regression	92.97%	36.86%	65.92%	33.07%	4.414%	93.27%
Random Forest	99.92%	82.42%	84.60%	81.71%	0.0464%	99.77%
Soft Voting Ensemble	99.93%	84.42%	87.67%	84.31%	0.0427%	99.98%
Stacking Ensemble	99.93%	86.35%	87.20%	84.78%	0.0406%	98.90%
XGBoost	99.91%	85.94%	89.27%	84.53%	0.0487%	99.98%

Note. All classifier models were evaluated after SMOTE-Tomek resampling across five validation folds.

The meaning of this result is that no classifier dominated every metric, but Stacking provided the most balanced trade-off across macro F1-score, precision, accuracy, and false-positive control. Stacking benefited from combining Random Forest, XGBoost, and Decision Tree base learners with Logistic Regression as a meta-learner, allowing the final model to use complementary decision patterns from different classifiers. This supports previous findings that voting and stacking ensembles can improve IDS robustness by reducing dependence on a single learner’s behavior (Al Essa & Bhaya, 2024; Emanet et al., 2023; Farooqi et al., 2024; Pramanick et al., 2025; Urmi et al., 2024). The final classifier choice is justified because macro F1-score reflects class-balanced performance more directly than accuracy, while macro FPR is important for avoiding excessive alerts in intrusion detection. The implication is that classifier selection should consider multiple macro metrics rather than one dominant metric. Consequently, the Stacking Ensemble was selected even though XGBoost and Soft Voting remained competitive in ROC-AUC.

The validation results also explain why the final model was not selected from accuracy alone. Logistic Regression produced the weakest macro F1-score, indicating that a linear decision surface was insufficient for this multiclass traffic distribution. KNN performed better than Logistic Regression but remained weaker than the tree-based and ensemble models, which is

consistent with the absence of feature scaling and the high sensitivity of distance-based methods to feature magnitude. XGBoost and Soft Voting were strong competitors, especially in ROC-AUC, but Stacking produced the highest macro F1-score and the lowest macro FPR among the evaluated classifier candidates. This combination made it the most defensible selection for a Brute-Force-centered NIDS pipeline.

After the selection stage, the final Stacking Ensemble with SMOTE-Tomek was trained on the combined train-validation partition and evaluated on 157,148 unseen test records. The final configuration achieved 99.93% macro accuracy, 86.10% macro precision, 90.22% macro recall, 86.93% macro F1-score, 0.0364% macro FPR, and 97.98% macro ROC-AUC. For Brute Force specifically, the model achieved 98.70% precision, 92.66% recall, 95.58% F1-score, and 0.0253% FPR across 3,187 test samples. The meaning of these values is that the selected pipeline detected most Brute Force samples while keeping false Brute Force alarms very low.

This result is stronger than the multiclass CICIDS2017 result reported by Khan et al. (2024), who used SMOTE-Tomek with conventional classifiers and reported 96.37% accuracy and 96.33% F1-score. The comparison is not fully one-to-one because the earlier study reported overall multiclass performance rather than the same Brute Force-specific test protocol. Nevertheless, the proposed model is justified as a stronger Brute-Force-oriented approach in this experimental setting because it combines train-only hybrid sampling, selected flow features, and heterogeneous stacking, and because it reports high Brute Force precision, recall, F1-score, and FPR on an unseen test set. The implication is that the proposed pipeline does more than improve general multiclass accuracy; it provides clearer evidence of Brute Force detection capability. Consequently, the model can be considered more suitable for Brute Force-focused NIDS evaluation on CICIDS2017 than the conventional-classifier approach reported in the earlier study.

The Brute Force result should be interpreted together with the macro results. A high Brute Force F1-score alone would not be sufficient if the model performed poorly on the other classes, because NIDS deployment requires decisions across many traffic categories. In this experiment, the selected model also achieved strong final test results for Benign, DDoS, Bot, and Port Scan traffic, while the weakest results were concentrated in the extremely small SQL-Injection, XSS, and Infiltrations classes. This pattern indicates that the pipeline improved the target Brute Force class without collapsing the broader multiclass task, but it also shows that resampling cannot fully overcome extremely limited original support.

Final Interpretation

Taken together, the results show that the proposed pipeline improved Brute Force detection through three connected mechanisms: SMOTE-Tomek increased minority-class representation, feature selection retained behaviorally meaningful flow attributes, and the Stacking Ensemble combined complementary classifiers into a stronger final decision model. This over-

all interpretation is supported by previous studies on hybrid sampling, feature selection, and ensemble learning in NIDS (Alsaffar et al., 2024; Farooqi et al., 2024; Khan et al., 2024; Nassreddine et al., 2025; Urmi et al., 2024). The pipeline is justified because each component addresses a specific weakness of imbalanced Brute Force detection: limited minority exposure, redundant flow attributes, and dependence on a single classifier. The practical implication is that Brute Force detection in CICIDS2017 benefits from treating imbalance handling, feature selection, and ensemble learning as one integrated workflow rather than as isolated steps.

The consequence is also important to interpret carefully. The final model performed strongly on Brute Force, Benign, DDoS, Bot, and Port Scan traffic, but extremely small classes remained less stable. SQL-Injection reached only 61.54% F1-score on six test samples, XSS reached 51.52% F1-score with low precision, and Infiltrations reached 90.00% F1-score on only 11 test samples. These outcomes suggest that SMOTE-Tomek helped the main Brute Force target more reliably than classes with extremely limited original support. Therefore, the results support the effectiveness of the selected pipeline on CICIDS2017, but broader deployment claims require validation on newer datasets and more diverse real-world traffic.

The final consequence is that the proposed approach should be viewed as a targeted improvement for Brute Force detection rather than a complete solution for all minority attacks. The method is effective when the minority class has enough original samples to support meaningful synthetic expansion and model learning. For classes with only a few original examples, synthetic balancing may create a larger training set, but it cannot fully replace real behavioral diversity. Future evaluation should therefore test whether the same pipeline remains effective when minority classes are represented by more varied traffic sources and when the dataset reflects newer attack environments.

This interpretation also explains why the final configuration was not selected solely from the strongest individual metric. XGBoost and Soft Voting showed competitive ROC-AUC values during validation, but the Stacking Ensemble offered the strongest macro F1-score and false-positive control among the classifier candidates. In an imbalanced NIDS context, this trade-off is important because operational usefulness depends on both detecting attacks and avoiding excessive incorrect alerts. The selected configuration therefore reflects a balanced decision rather than a metric-specific optimization.

CONCLUSION AND FUTURE RESEARCH

This study presented a multiclass NIDS pipeline for Brute Force detection using CICIDS2017. The pipeline combined label consolidation, Benign-class limitation, train-only feature selection, SMOTE-Tomek hybrid sampling, and ensemble learning. The main purpose was to evaluate whether these components could work together to improve Brute Force detection under class imbalance without allowing validation or test data to influence training decisions.

The results show that this integrated design is useful because each component addresses a different part of the problem: feature selection reduces redundant flow attributes, SMOTE-Tomek improves minority-class representation, and stacking combines complementary classifiers.

SMOTE-Tomek substantially improved Brute Force training representation from 5,204 to 76,385 samples and reduced the imbalance against DDoS from about 30.30:1 to 2.06:1. The sampling-selection results showed that SMOTE-Tomek achieved the strongest macro F1-score, macro precision, macro recall, and macro ROC-AUC among the evaluated sampling strategies. After validation-based classifier selection, the Stacking Ensemble with SMOTE-Tomek was chosen as the final configuration because it produced the highest validation macro F1-score while maintaining low macro FPR.

On the unseen test set, the selected configuration achieved 99.93% macro accuracy, 86.10% macro precision, 90.22% macro recall, 86.93% macro F1-score, and 97.98% macro ROC-AUC. For Brute Force traffic, it achieved 98.70% precision, 92.66% recall, 95.58% F1-score, and 0.0253% FPR. These results indicate that hybrid sampling and heterogeneous ensemble learning can improve Brute Force detection under multiclass imbalance when they are applied within a leakage-aware training workflow. The comparison with previous CICIDS2017 work also suggests that the proposed configuration provides stronger Brute-Force-oriented evidence than conventional classifiers using SMOTE-Tomek alone, although the comparison should be interpreted carefully because evaluation protocols differ.

Several limitations remain. First, the experiment used only CICIDS2017, so the results may not generalize to newer or different network environments, which is a known concern in ML-based NIDS evaluation (Gu et al., 2025; Sarhan et al., 2021; Wardana et al., 2024). Second, SMOTE-Tomek helped the main Brute Force target more reliably than extremely small classes. SQL-Injection was evaluated on only six test samples, XSS still had weak precision, and Infiltrations had only 11 test samples. Third, synthetic resampling may introduce bias when original support is very limited. Fourth, exact package versions from the Colab runtime were not recorded. Finally, no feature scaling or normalization was applied, which may affect distance-based and linear components such as KNN and Logistic Regression during validation.

Future work should validate the pipeline on additional datasets and newer traffic environments to test whether the selected features and classifier behavior remain stable beyond CICIDS2017. Further experiments should also record exact dependency versions, evaluate train-only feature scaling for models that are sensitive to feature magnitude, and compare the proposed workflow against additional ensemble configurations under the same leakage-aware protocol. For extremely rare classes, future studies should investigate whether additional real samples, alternative resampling strategies, or few-shot approaches can improve detection without over-relying on synthetic data. These directions would help determine whether the proposed Brute-Force-centered pipeline can be extended into a more general minority-attack detection framework.

Overall, the study demonstrates that improving minority-attack detection requires more

than adding a resampling method to an existing classifier. The strongest result emerged when class representation, feature relevance, and model diversity were addressed together. This conclusion is useful for future NIDS research because it emphasizes the need to evaluate the full modeling workflow, from data preparation to final class-wise testing, rather than treating pre-processing and classification as separate concerns.

In practical terms, the findings suggest that Brute Force detection should be evaluated with attention to both missed attacks and false alarms. High recall is important because undetected Brute Force traffic can allow repeated unauthorized access attempts to continue, while high precision and low FPR are important because excessive false alarms can reduce trust in NIDS alerts. The selected Stacking Ensemble with SMOTE-Tomek achieved a strong balance between these goals on CICIDS2017. This balance is the main reason the proposed workflow is considered appropriate for Brute-Force-focused evaluation, even though additional datasets are still needed before broader deployment claims can be made.

Thus, the contribution of this study is not only the final score, but also the evidence that each stage of the pipeline supports the Brute Force detection objective.

OPEN DATA

The CICIDS-2017 dataset is publicly available via the CIC repository and can be obtained from the following website: <https://www.unb.ca/cic/datasets/ids-2017.html>. The source code used in this study is publicly available via GitHub and can be obtained from the following website: https://github.com/jetasunanda/Enhancing_Brute_Force_Attack_Detection.

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